

Practical: 6

Implementation of chain matrix multiplication using dynamic programming

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Aim: To Implement the chain matrix multiplication using dynamic programming

Algorithm:

Practical 6

Algorithm :

input : n matrices and dimension array,
 $P[0 \dots n]$

Output : minimum no. of scalar multiplications
required to multiply all matrices

Steps :

1. Start
2. read the number of matrices n
3. read the dimension array
 $P[0], P[1] \dots P[n]$
4. Create two tables
 $E[n+1][n+1]$
 $K[n+1][n+1]$
5. Initialize the cost table for a single matrix
 $C[i][j] = 0$
6. calculate the ending matrix
 $d = 1$ to $n-1$ repeat for loop
 $j = i + d$
7. for $x = i$ where $x < j$ do
 $q = C[i][x] + C[x+1][j] + P[i-1] * P[x] * P[j]$
8. if $(min > q)$ then do
 $min = q$
 $K[i][j] = x$
9. increment the x values
10. Assign $C[i][j] = min$
11. stop

Program:

```
#include <iostream>
using namespace std;

int main(){
    int n = 5;

    int p[] = {5, 10, 3, 12, 5, 50};

    int c[10][10] = {0};
    int k[10][10] = {0};

    for(int d = 1; d < n; d++){
        for(int i = 1; i <= n-d; i++){
            int j = i + d;
            int min = 99999;

            for(int x = i; x < j; x++){
                int q = c[i][x] + c[x+1][j]
                    + p[i-1] * p[x] * p[j];

                if(q < min){
                    min = q;
                    k[i][j] = x;
                }
            }
            c[i][j] = min;
        }
    }

    cout << "\nMinimum cost = " << c[1][n]<<endl;

    return 0;
}
```

Output:

Output

Minimum cost = 1655

Time Complexity:

Time Complexity:

```

int n=5;
int p[] = {5, 10, 3, 12, 5, 80};
int c[10][10] = {0};
int k[10][10] = {0};
for (int d=1; d<n; d++) {
    for (int i=1; i<=n-d; i++) {
        int j=i+d;
        int min=99999;
        for (int x=i; x<j; x++) {
            int q = c[i][x] + c[x+1][j] + p[i+1] * p[x+1] * p[j];
            if (q < min) {
                min = q;
                k[i][j] = x;
            }
        }
        c[i][j] = min;
    }
}

```

$$T(n) = \sum_{d=1}^{n-1} (n-d) \cdot d$$

$$T(n) = \sum_{d=1}^{n-1} (nd - d^2)$$

$$T(n) = n \sum_{d=1}^{n-1} d - \sum_{d=1}^{n-1} d^2$$

$$T(n) = n \cdot \frac{n(n-1)}{2} - \frac{n(n-1)(2n-1)}{6}$$

$$T(n) = \frac{n^3 - n^2}{2} - \frac{2n^3 - 3n^2 + n}{6}$$

$$T(n) = \frac{3n^3 - 3n^2 - 2n^3 + 3n^2 - n}{6}$$

$$T(n) = \frac{n^3 - n}{6}$$

$$\Rightarrow T(n) = O(n^3)$$

Best case : $O(n^2)$
 Average case : $O(n^3)$
 Worst case : $O(n^3)$

Conclusion:

The Implementation of the of chain matrix multiplication using dynamic programming has successfully executed with the time complexity of $O(n^3)$ in the Three cases of best, average and worst. At the end of this program says that the Minimum number of scalar multiplications required to multiply all matrices.